Joint-Embedding Predictive Architectures and Latent Residual Conditioning for Multivariate Time-Series Anomaly Detection

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***Abstract*—Deep reconstructive detectors for multivariate time-series anomaly detection minimize point-wise error in raw observation space. Because sensor telemetry contains unforecastable jitter and scale disparities, observation-space residuals develop heavy tails that trigger severe false-alarm flooding under automated threshold calibration. Joint-Embedding Predictive Architectures (JEPAs) avoid reconstructing raw noise by forecasting compact latent summaries of future windows. We investigate how latent prediction conditions residual distributions under regime shift. Specifically, we compare a single-horizon baseline (TS-JEPA) against Soliton-Reynolds JEPA, a multi-horizon architecture that taps intermediate temporal convolutional stages, subtracts online context barycenters, whitens multi-horizon prediction errors, and scores centered sub-horizon fluctuations via a Lieb–Thirring spectral concentration ratio. Across continuous physical telemetry benchmarks calibrated on held-out nominal data with Streaming Peaks-Over-Threshold (SPOT), latent scoring suppresses false alarms under both retrospective buffered attribution and strictly causal streaming while improving anomaly ranking and event recall. In addition, replacing scalar score aggregation with a 224-bit fluid-dynamic circuit-breaker panel that monitors sub-horizon positions, Reynolds slow-drift transport, turbulent fluctuations, and right-edge convective acceleration with fast onset relays prevents isolated drifting coordinates from inflating the alarm threshold, nearly doubling causal point-wise F1 and cutting detection delay in half across 141 multivariate streams. Controlled factorial ablations also show that layer-normalization bottlenecks compress residual kurtosis, learned latent projections and barycenter whitening curb calibration-to-holdout drift, short prediction horizons eliminate emission lag, and hybrid sensor-variance monitors recover quiet single-channel faults.**

***Index Terms*—Time-series anomaly detection, Multivariate analysis, Threshold calibration, Extreme value theory, Representation learning.**

I. Introduction

Autonomous anomaly detection in multi-sensor telemetry monitors operational health across aerospace systems, Earth observation, cyber-physical infrastructure, and wearable healthcare [1]–[3]. In NASA SMAP and MSL rovers, industrial process loops, and biometric sensor arrays, dozens of channels track power buses, thermal gradients, vibrations, and kinematics [4]. Monitors must identify subsystem faults without overwhelming operators with nuisance alerts.

False-Alarm Flooding Under Automated Threshold Calibration: In deployed monitoring, an automated rule must convert continuous anomaly scores into binary alarms using healthy (nominal) calibration data alone. Standard deep detectors (TimesNet, TranAD, Anomaly Transformer) reconstruct raw sensor readings and flag timestamps with large point-by-point reconstruction error [5]–[7]. Because raw telemetry contains sensor jitter and scale differences across channels, point-level errors produce frequent, clustered spikes and a heavy-tailed residual distribution with high excess kurtosis ($\gamma_2 > 8$, Table I). Under Streaming Peaks-Over-Threshold (SPOT) [8] at a 0.1% target risk ($q = 10^{-3}$), normal background shifts easily cross the threshold, triggering false alarms on 22%–31% of healthy timesteps.

Latent Residual Conditioning and Scope of Claims: Predicting a compact *summary* of a future time window rather than every noisy sensor sample filters out random jitter before the prediction error is computed. We study Joint-Embedding Predictive Architectures (JEPAs) [9], which compress the preceding 256 timesteps into a 32-dimensional summary vector and forecast the summary of the next S timesteps ($S = 64$ by default, or $S \in \{1, 4\}$ for fast tripwires) using healthy training data alone [10]. Alongside a single-horizon baseline (TS-JEPA), we introduce *Soliton-Reynolds JEPA*, which taps multiple depths of its temporal convolutional network [11], subtracts the running context barycenter to cancel slow level shifts, and whitens prediction errors across $M = 4$ nested sub-horizons. Drawing on Reynolds decomposition [12] and Lieb–Thirring spectral bounds [13], it separates multi-horizon errors into mean transport and zero-mean fluctuations, measuring whether fluctuation energy is dispersed like background noise or concentrated into a coherent rank-1 wavepacket. We also resolve a critical failure mode of scalar score aggregation: collapsing 128 whitened residuals into a single 1D score before calibration allows one or two mildly drifting coordinates (*hijacker dials*) to inflate the SPOT threshold and suppress causal Point-F1. Replacing 1D collapse with a *224-bit multi-perspective circuit-breaker panel* (monitoring seven fluid-dynamic dials per coordinate with independent Boolean tripwires and 32 fast right-edge onset relays) eliminates hijacker inflation without test-set tuning. We evaluate four questions: (i) how much latent residuals and Boolean circuit breakers lower realized false-positive rates (FPRs) and calibration error $|\text{FPR}_{\text{hold}} - q|$;

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(ii) how normalization, learned latent targets, multi-horizon whitening, and wavepacket concentration each contribute; (iii) how detection varies across point spikes, contextual faults, and quiet single-sensor freezes; and (iv) how shortening S trades off alarm lag ($S-1$) against detection quality.

A. Contributions

1) *Why latent predictors stabilize alarm thresholds:* Across 12 matched-backbone models (272k–312k online parameters), a layer-normalization (LayerNorm) [14] bottleneck compresses residual kurtosis from 6.10 to 0.91, while predicting a learned latent summary cuts calibration-to-holdout drift in half (D_{KS} from 0.376 to 0.214 in TS-JEPA) and multi-horizon barycenter whitening in Soliton-Reynolds JEPA lowers drift further to $D_{\mathrm{KS}} = 0.179$ (Section VI-D, Table II(c)).
2) *Fewer false alarms, contextual-anomaly dominance, and quiet-fault recovery:* Across the 19 non-GHL streams, latent scoring cuts held-out FPRs three-fold under buffered attribution (7.8%–9.3% vs. 30.0%–30.7% for TimesNet and TranAD) and by over an order of magnitude causally (0.6%–2.4% vs. 22.2%–24.3%). Soliton-Reynolds JEPA matches TS-JEPA on buffered precision-recall area under the curve (PR-AUC 0.331), catches 56.2% of anomaly events at a 1% false-alarm budget, and leads on NASA contextual anomalies (0.652 buffered PR-AUC, 75.0% causal SPOT recall at 0.6% FPR, and 82.0%–92.0% recall at 1%–5% FPR budgets), while a rolling sensor-variance check recovers quiet single-channel freezes (Tables II(c), III, and IV, Figs. 3(b) and 4(b)).
3) *Multi-horizon wavepacket bounds, 224-bit circuit-breaker panel, and fast response:* Proposition 1 proves that the Reynolds-centered Lieb–Thirring concentration ratio satisfies $\frac{1}{M-1} \le R_{\mathrm{LT}} \le 1$ with exact radial scale invariance. While post-hoc wavepacket multipliers satisfy Two One-Sided Tests (TOST) [15] equivalence, replacing 1D scalar collapse with the 224-bit circuit-breaker panel more than doubles causal Point-F1 (0.052 → 0.116–0.138 on n=19 streams, 17/0 wins, $p = 1.5\times10^{-5}$; 0.067–0.077 → 0.125–0.139 across n=141 streams, 114/18 wins, $p = 1.2\times10^{-16}$) and halves detection delay. Short horizons ($S \in \{1, 4\}$) eliminate the 63-step window lag (Tables II and III, Fig. 4(a)).
4) *Why automated thresholds fail and when they hold:* Permutation decomposition with bootstrap confidence intervals (CIs) shows that slow regime drift causes 71.3% of excess causal false alarms in TS-JEPA and 78.3%–81.8% in reconstructors. Barycenter centering in Soliton-Reynolds JEPA cuts this inter-block drift share four-fold to 18.7%, while runs-declustered SPOT and rolling quantiles stabilize thresholds under step and variance shifts (Table V).

Roadmap. Section II reviews related work, Section III details the benchmark, Section IV formulates the architectures, Section V defines the protocol, Sections VI–VII report results, and Section VIII concludes.

TABLE I
KEY OPERATIONAL AND STATISTICAL TERMS.

Term	Meaning
Latent residual	Gap between the predicted and observed 32-dimensional summary of the next window.
SPOT	Streaming extreme-value rule that sets an alarm threshold from healthy calibration scores alone.
224-bit breaker	Boolean panel of 7×32 fluid-dynamic coordinate tripwires and 32 onset relays avoiding 1D score collapse.
Calibration error	Gap $\lvert\mathrm{FPR}_{\mathrm{hold}} - q\rvert$ between the actual false-alarm rate on healthy test data and the 0.1% target (q=10^{-3}).
Event recall	Fraction of true anomaly segments with at least one alarm.
Buffered / Causal	Retrospective S-step window averaging vs. real-time trailing attribution at $\tau = t + S - 1$.
Calibration drift	Shift between calibration and later healthy residuals, measured by Kolmogorov–Smirnov distance.
Excess kurtosis	Tail heaviness of the residual distribution; large values mean frequent, extreme spikes.
Soliton ratio	Score $R_{\mathrm{LT}} \in [\frac{1}{M-1}, 1]$ measuring whether sub-horizon errors align into one coherent wavepacket.

II. RELATED WORK

Observation-Space Forecasting and Reconstructive Detectors: Deep time-series anomaly detectors predominantly forecast sensor readings or reconstruct input windows in raw observation space, flagging timestamps where point-level error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ exceeds a threshold. Autoregressive long short-term memory (LSTM) networks with non-parametric dynamic thresholding [4] model sequential dependencies directly on spacecraft channels. Recent Transformer and spectral reconstructors introduce specialized attention and frequency modules: TranAD [7] combines adversarial two-phase training with self-conditioning sub-series attention; Anomaly Transformer [5] contrasts prior Gaussian kernel associations against series-level self-attention; and TimesNet [6] folds 1D sequences into 2D tensors along dominant Fast Fourier Transform (FFT) periods for 2D convolutional modeling. Because point-level losses penalize error across every channel equally, reconstructors fit stochastic sensor jitter and environmental fluctuations. In high-dimensional streams ($K \ge 25$), summing raw channel errors accumulates heavy-tailed (*leptokurtic*, $\gamma_2 > 0$) residual spikes that destabilize extreme-value threshold fitting under baseline shift [16].

Contrastive and Joint-Embedding Predictive Architectures: Self-supervised representation methods map windows into a compact latent space [17]. Contrastive detectors construct positive and negative pairs: temporal and contextual contrasting [18] and TS2Vec [19] maximize contextual agreement across hierarchical sub-series; Neural Contextual Anomaly Detection (NCAD) [10] contrasts full-window and context-only embeddings using injected synthetic anomalies; and DCdetector [20] employs dual patch-wise and in-patch attention without reconstruction. However, contrastive detectors depend on whether synthetic perturbations or patch discrepancies match test-time fault morphology (0.058 and 0.070 macro Point-F1 for NCAD and DCdetector vs. 0.160 for latent prediction across 11 datasets, Table III(d)). Joint-Embedding Predictive Architectures (JEPAs) [9], [21], [22] eliminate observation decoding and negative pairs by encoding context z_{ctx} and predicting the target embedding z_{tgt} produced by a separate target encoder. Collapse is prevented via exponential moving average (EMA) target updates [23] or variance-covariance penalties such as VICReg [24], Barlow Twins [25], and batch whitening [26]. How latent predictive residuals behave under unsupervised extreme-value threshold calibration remains unexplored.

Fluid-Dynamic Reynolds Decomposition and Spectral Concentration Priors: Multi-sensor telemetry reflects physical dynamical trajectories [27]. Neural ordinary differential equations (ODEs) [28], Physics-Informed Neural Networks (PINNs) [29], and Hamiltonian or Lagrangian networks [30], [31] enforce differential conservation laws on physical coordinates, but require known governing equations and suffer from stiff gradient pathologies on uncalibrated sensor arrays [32]. Koopman and operator-theoretic models [33]–[37] lift nonlinear dynamics into learned observable spaces. Rather than imposing partial differential equations on raw sensors, we draw on Reynolds decomposition [12] from turbulence modeling to separate multi-horizon latent predictions into a mean trajectory and centered sub-horizon fluctuations. We then measure wavepacket localization across prediction horizons via a Lieb–Thirring spectral concentration ratio [13] on the centered fluctuation Gram matrix.

Extreme-Value Threshold Calibration and Evaluation Rigor: Grounded in the Pickands–Balkema–de Haan extreme-value theorem [38], Streaming Peaks-Over-Threshold (SPOT) [8] fits a Generalized Pareto Distribution (GPD) to score exceedances above quantile u to extrapolate a threshold at design risk q. However, continuous telemetry violates the classical independent and identically distributed (i.i.d.) assumption through short-range serial correlation (quantified by the Ferro–Segers extremal index [39]) and inter-block regime drift. Because "point-adjusted" F1 severely inflates scores [16], [40], [41], we evaluate all models using unadjusted Point-F1, PR-AUC, causal vs. buffered attribution, and matched realized false-alarm operating curves.

III. Multi-Sensor Internet-of-Things and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams from NASA spacecraft archives [4], the University of California, Irvine (UCI) Machine Learning Repository [42], [43], TimeEval [16], and mTSBench [44]. Thirty-three streams enter the primary multi-seed held-out protocol; 12 are reserved with explicit accounting below. The suite spans four physical domains and separates natural operational or physiological anomalies from testbed or simulated faults. *Spacecraft and solar telemetry* covers six SMAP streams (K=25 channels), six MSL rover streams (K=55, four primary and two short-nominal), and SWAN-SF solar magnetograms [45] (K=38), all containing natural operational or solar-flare events; on SMAP and MSL, NASA's ground-truth archive [4] further classifies each anomaly sequence as either an out-of-range *point* fault or an in-range *contextual* fault (Table IV(a)). *Wearable kinematics* covers six Daphnet Freezing-of-Gait (FoG) streams [42] sampled at 64 Hz (K=9, 72 natural gait-freezing episodes, where S=64 equals a 1 s horizon). *Industrial processes and smart infrastructure* covers 14 Gasoil Heating Loop (GHL) streams (K=16, 26 testbed valve and heater faults), the Genesis pick-and-place testbed (K=18), GECCO water chemistry at 1-minute resolution (K=9, 37 natural hydrological anomalies), Metro I-94 hourly highway traffic (K=5), Room-Occupancy sensing [46] (K=5), CalIt2 building flow [43] (K=2), and six CICIDS network streams (K=72, simulated attack flows). Cohort 3 (Section VI-B) separately injects six controlled single-channel faults into six real streams.

Each stream provides nominal training sequences ($N_{\text{tr}} \in [2{,}074, 120{,}000]$ across the 33 primary streams) and labeled anomaly test sequences ($N_{\text{test}} \in [2{,}158, 114{,}001]$). Channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2} against flatlines, matching standardized archive subsets in prior benchmarks [4], [16].

Each nominal sequence is split chronologically into training (65%), nominal calibration (17.5%), and held-out nominal evaluation (17.5%) blocks separated by a 64-step isolation gap (Section V-A). Primary 33-stream inclusion required $N_{\text{cal}} \geq 500$ ($T_{\text{nom}} \geq 3{,}769$) and $K \leq 55$, $N_{\text{test}} \leq 1.2 \times 10^5$ (Table III(c)); among the 12 reserved streams, five have $T_{\text{nom}} \in [764, 1{,}492]$ ($N_{\text{cal}} \leq 101$ across CalIt2, MSL C-2, MSL D-16, and two Room-Occupancy streams), six CICIDS streams have K=72, $N_{\text{test}} \geq 1.36 \times 10^5$, and GECCO ($K$=9, 1 held-out event) enters Cohorts 3–4. To eliminate subset-selection bias, Cohort 5 evaluates all 141 qualifying multivariate streams across 12 datasets ($N_{\text{cal}} \geq 100$ and ≥ 1 test anomaly: 47 SMAP, 22 Daphnet, 18 MSL, 18 SMD, 17 Exathlon, 13 Opportunity, GECCO, Genesis, Pooled Server Metrics (PSM), credit-card, Metro, and SWAN) at three seeds alongside the 130-stream baseline screen (Table III(d)).

Two caveats bound external validity: (i) SMAP and MSL contain documented labeling flaws [41]; and (ii) streams within SMAP, MSL, Daphnet, and GHL share hardware platforms (N=7 dataset clusters). We therefore report Dataset-Macro means, between-stream and within-stream seed standard deviations (SDs), zero-tuning leave-one-dataset-out (LODO) sensitivity, and dataset-clustered permutation tests (Table III(c)).

IV. Methodology: Joint-Embedding Predictive Architectures and Soliton-Reynolds Decomposition

A. Problem Formulation and Latent Space Prediction

Consider a multivariate telemetry stream $\mathbf{X} \in \mathbb{R}^{T \times K}$ recording K sensor channels over T timesteps. Fig. 1 illustrates the four-stage pipeline from raw telemetry to calibrated binary alarms. In Stage 1, a sliding window of length $W = C + S = 320$ advances at stride 1 and splits the series at index t into a *context window* $\mathbf{x}_{\text{ctx}} = \mathbf{X}[t-C : t, :] \in \mathbb{R}^{C \times K}$ ($C = 256$) and a *suspect target horizon* $\mathbf{x}_{\text{tgt}} = \mathbf{X}[t : t+S, :] \in \mathbb{R}^{S \times K}$ ($S = 64$, up to decision timestamp $\tau = t + S - 1$). Each window passes through a 4-shell dyadic real fast Fourier transform (rFFT) front-end and a causal dilated temporal convolutional network (TCN) [11].

Unlike reconstructive detectors that minimize pointwise observation error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ and inherit leptokurtic sensor noise (Sections I–II), Stage 2 shifts prediction into a 32-dimensional latent space ($D = 32$) using a Joint-Embedding Predictive Architecture (JEPA) [9]. Dual online (E_θ) and target (E_ϕ) encoders extract three multi-depth taps across the shallow, mid, and deep TCN hierarchy and map $\mathbf{x}_{\text{ctx}}$ and $\mathbf{x}_{\text{tgt}}$ into

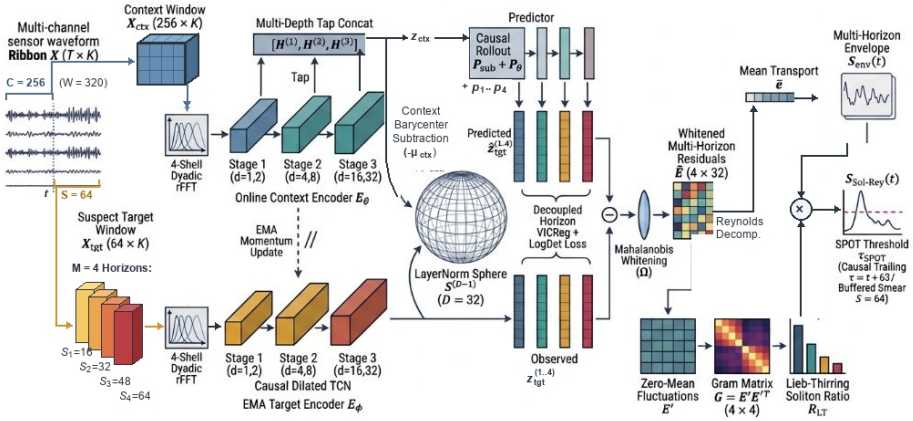


Fig. 1. Overview of the four-stage Soliton-Reynolds JEPA pipeline: (1) multiscale windowing and causal encoding, (2) dual-encoder hierarchy tapping and barycenter centering, (3) nested multi-horizon latent rollout, and (4) whitened residual scoring and extreme-value or 224-bit circuit-breaker calibration.

spherical representations $z_{\text{ctx}}, z_{\text{tgt}} \in \sqrt{D}\mathbb{S}^{D-1}$. Target weights ϕ are updated as an exponential moving average (EMA) of online weights θ, $\phi_k \leftarrow m_k\phi_{k-1} + (1-m_k)\theta_k$ with cosine momentum $m_k \in [0.995, 0.9995]$ [21], [23]. Stage 3 rolls out latent predictions across $M = 4$ nested sub-horizons ($S_m \in \{16, 32, 48, 64\}$) under a two-term multi-horizon mean-squared error (MSE) and covariance whitening defect objective (Section IV-B), and Stage 4 applies Mahalanobis residual whitening, Lieb–Thirring soliton concentration modulation, and either coordinate-saliency-gated SPOT calibration or the 224-bit multi-perspective circuit-breaker panel under causal trailing ($\tau = t + 63$) and buffered smear mappings (Section IV-C).

B. Latent Predictor Architectures: TS-JEPA and Soliton-Reynolds JEPA

We evaluate two nested latent predictors sharing the same multiscale encoder backbone and coordinate saliency gate: an unconstrained single-horizon baseline (TS-JEPA) and a multi-horizon fluid-dynamic decomposition model (Soliton-Reynolds JEPA).

1) Unconstrained TS-JEPA Baseline: In TS-JEPA, a two-layer multi-layer perceptron (MLP) predictor $P_\theta : \mathbb{R}^D \to \mathbb{R}^D$ maps z_{ctx} to a single full-horizon prediction $\hat{z}_{\text{tgt}} = P_\theta(z_{\text{ctx}})$ trained with standard single-horizon predictive alignment and VICReg variance-covariance penalties [24] ($\gamma_{\text{var}} = 1.0$). Terminal LayerNorm [14] bounds latent representations on $\sqrt{D}\mathbb{S}^{D-1}$ and compresses nominal residual kurtosis relative to observation-space reconstructors, but single-horizon encoding pools the entire $S = 64$ target window into one terminal TCN state, smoothing out short sub-horizon transients.

2) Soliton-Reynolds JEPA (Multi-Horizon Reynolds Decomposition and Lieb–Thirring Concentration): Physical Motivation and Structural Design: In fluid turbulence, Reynolds decomposition [12] separates a velocity field into its time-averaged mean flow and zero-mean fluctuations, while dispersive wave theory characterizes localized wavepackets (*solitons*) by spectral concentration of the fluctuation operator [13]. Soliton-Reynolds JEPA incorporates both principles through two stages before residual scoring:

1) *Multi-depth TCN hierarchy taps and synchronized barycenter centering:* To capture sharp onset transients attenuated by deep dilated blocks, a multi-depth tap head extracts $L_{\text{tap}} = 3$ feature maps $\mathbf{H}^{(\ell)} \in \mathbb{R}^{B\times T_w \times d_{\text{model}}}$ ($d_{\text{model}} = 48$) across the shallow dyadic front-end, mid-depth block, and terminal TCN block. For each tap, the centered terminal state and temporal root-mean-square (RMS) energy are concatenated into $\mathbf{h}_{\text{tap}} \in \mathbb{R}^{B\times 6d_{\text{model}}}$, mapped into $\mathbb{R}^D$, and fused with the deep bottleneck via gate $g = 0.5\,\sigma(\theta_g) \le 0.5$. Subtracting the online running context barycenter $\bar{\boldsymbol{\mu}}_{\text{ctx}}$ from both online context and EMA target fused embeddings prior to LayerNorm projection onto $\sqrt{D}\mathbb{S}^{D-1}$ cancels common-mode baseline drift.
2) *Nested causal multi-horizon rollout and covariance whitening defect regularization:* We partition $\mathbf{x}_{\text{tgt}}$ into $M = 4$ nested causal prefixes $\mathbf{x}_{\text{tgt}}^{(m)} = \mathbf{x}_{\text{tgt}}[0 : S_m, :]$ ($S_m = (m/M)S \in \{16, 32, 48, 64\}$), encoded by E_ϕ into $z_{\text{tgt}}^{(m)} \in \mathbb{R}^D$. From z_{ctx}, sub-horizon predictor P_{sub} rolls out $\hat{z}_{\text{tgt}}^{(m)} = \text{LN}\big(P_{\text{sub}}(\hat{z}_{\text{tgt}}^{(m-1)} + \mathbf{p}_m)\big)$ for $m < M$ ($\hat{z}_{\text{tgt}}^{(0)} \equiv z_{\text{ctx}}$) using horizon embeddings $\mathbf{p}_m \in \mathbb{R}^D$, while P_θ forecasts $\hat{z}_{\text{tgt}}^{(M)} = \text{LN}\big(P_\theta(z_{\text{ctx}} + \mathbf{p}_M)\big)$. With $w_m = \frac{1}{2(M-1)}$ ($m < M$) and $w_M = \frac{1}{2}$, training minimizes multi-horizon predictive MSE plus the Frobenius covariance whitening defect:

$$\mathcal{L}_{\text{Sol-Rey}} = \sum_{m=1}^{M} \frac{w_m}{D}\Big(\big\|\hat{z}_{\text{tgt}}^{(m)} - \text{sg}\big(z_{\text{tgt}}^{(m)}\big)\big\|_2^2 + \big\|\boldsymbol{\Delta}_\gamma\big(\hat{z}_{\text{tgt}}^{(m)}\big)\big\|_F^2\Big), \tag{1}$$

where $\boldsymbol{\Delta}_\gamma(\mathbf{Z}) \in \mathbb{R}^{D\times D}$ is the whitening defect of centered batch predictions $\mathbf{Z}_c = \mathbf{Z} - \mathbf{1}_B\bar{\mathbf{z}}^\top$ with covariance $\boldsymbol{\Sigma}(\mathbf{Z}) = (B-1)^{-1}\mathbf{Z}_c^\top\mathbf{Z}_c$ and standard deviations $\sigma_d(\mathbf{Z}) = \sqrt{\Sigma_{dd}(\mathbf{Z}) + \epsilon}$ ($\gamma = 0.75$, $\epsilon = 10^{-4}$), defined by $[\boldsymbol{\Delta}_\gamma(\mathbf{Z})]_{dd} = \sqrt{\max(0, \gamma - \sigma_d(\mathbf{Z}))}$ and $[\boldsymbol{\Delta}_\gamma(\mathbf{Z})]_{dj} = \frac{1}{\sqrt{2}}\Sigma_{dj}(\mathbf{Z})$ ($d \ne j$). Unlike log-determinant barriers that fail when $B = 16 < D = 32$, $\|\boldsymbol{\Delta}_\gamma(\mathbf{Z})\|_F^2$ remains smooth for any $B \ge 2$ and raises effective rank $\text{RankMe}(z_{\text{tgt}})$ from 11.6 to 19.8 [47].

Reynolds Fluctuation Gram Matrix and Lieb–Thirring Soliton Concentration: Let $\mathbf{E} = [\mathbf{e}_1, \ldots, \mathbf{e}_M]^\top \in \mathbb{R}^{M\times D}$ stack the M prediction residuals $\mathbf{e}_m = z_{\text{tgt}}^{(m)} - \hat{z}_{\text{tgt}}^{(m)}$. Fitting the shrinkage precision matrix $\boldsymbol{\Omega} = (\hat{\boldsymbol{\Sigma}}_{\text{res}} + 10^{-3}\mathbf{I}_{MD})^{-1/2} \in \mathcal{S}_{++}^{MD}$ [48] and mean $\bar{\boldsymbol{\mu}}_{\text{res}} \in \mathbb{R}^{MD}$ on nominal training windows yields Mahalanobis-whitened residuals $\tilde{\mathbf{e}} = \boldsymbol{\Omega}(\text{vec}(\mathbf{E}) - \bar{\boldsymbol{\mu}}_{\text{res}}) \in \mathbb{R}^{MD}$, reshaped into $\tilde{\mathbf{E}} = [\tilde{\mathbf{e}}_1, \ldots, \tilde{\mathbf{e}}_M]^\top \in \mathbb{R}^{M\times D}$. Reynolds decomposition across horizons separates $\tilde{\mathbf{E}}$ into horizon-mean transport $\bar{\mathbf{e}} = M^{-1}\sum_{m=1}^{M}\tilde{\mathbf{e}}_m \in \mathbb{R}^D$ and zero-mean fluc-

tuations $\mathbf{E}' = \tilde{\mathbf{E}} - \mathbf{1}_M \bar{\mathbf{e}}^\top \in \mathbb{R}^{M\times D}$. From the fluctuation Gram matrix $\mathbf{G} = \mathbf{E}'\mathbf{E}'^\top \in \mathcal{S}_+^M$ with ordered eigenvalues $0 \le \lambda_1 \le \cdots \le \lambda_M$, the Lieb–Thirring spectral concentration ratio [13] of order $\gamma = 1$ is:

$$R_{\text{LT}}(\mathbf{E}') = \frac{\lambda_M^{\gamma+1/2}}{\sum_{m=1}^{M} \lambda_m^{\gamma+1/2} + \epsilon_{\text{LT}}}, \quad \epsilon_{\text{LT}} = 10^{-12}. \tag{2}$$

Proposition 1 (Lieb–Thirring Soliton Concentration Bounds, Centered Gram Rank Deficiency, and Scale Invariance)**.** *Let $\mathbf{E}' = (\mathbf{I}_M - M^{-1}\mathbf{1}_M\mathbf{1}_M^\top)\tilde{\mathbf{E}} \in \mathbb{R}^{M\times D}$ denote the Reynolds-centered multi-horizon fluctuation matrix for $M \ge 2$ and $\gamma > -1/2$. In the exact limit $\epsilon_{LT} \to 0^+$ for any non-zero fluctuation $\mathbf{E}' \ne \mathbf{0}$: (i)* Centered Gram nullspace: *Because $\mathbf{E}'^\top\mathbf{1}_M = \mathbf{0}$, $\mathbf{1}_M \in \ker(\mathbf{G})$, forcing $\lambda_1(\mathbf{G}) \equiv 0$ and* $\operatorname{rank}(\mathbf{G}) \le \min(M-1, D)$. *(ii)* Sharp isotropy and soliton bounds: *The concentration ratio satisfies*

$$\frac{1}{M-1} \le R_{LT}(\mathbf{E}') \le 1, \tag{3}$$

attaining $\frac{1}{M-1}$ ($= \frac{1}{3}$ for $M = 4$) iff fluctuation energy is distributed isotropically across $M-1$ modes ($\lambda_2 = \cdots = \lambda_M > 0$), and 1 iff $\operatorname{rank}(\mathbf{E}') = 1$ *($\lambda_2 = \cdots = \lambda_{M-1} = 0 < \lambda_M$). (iii)* Exact radial scale invariance: *For any $c > 0$, $R_{LT}(c\mathbf{E}') = R_{LT}(\mathbf{E}')$.*

Proof. Since $\mathbf{G}\mathbf{1}_M = \mathbf{E}'(\mathbf{E}'^\top\mathbf{1}_M) = \mathbf{0}$, $\lambda_1 = 0$, leaving at most $M-1$ positive eigenvalues $\lambda_2 \le \cdots \le \lambda_M$ with $\lambda_M > 0$. Monotonicity of $x \mapsto x^{\gamma+1/2}$ on $x \ge 0$ yields $\lambda_M^{\gamma+1/2} \le \sum_{m=2}^{M} \lambda_m^{\gamma+1/2} \le (M-1)\lambda_M^{\gamma+1/2}$; dividing by the sum gives (3) with the stated equality conditions. Rescaling $\mathbf{E}' \mapsto c\mathbf{E}'$ scales $\lambda_m^{\gamma+1/2} \mapsto c^{2\gamma+1}\lambda_m^{\gamma+1/2}$, which cancels in (2) as $\epsilon_{\text{LT}} \to 0^+$. □

C. Anomaly Scoring and Operational Detection Latency

At test time, windows are scored at stride 1. Let $g_d : \mathbb{R}^d \to \mathbb{R}_{\ge 0}$ denote the coordinate saliency gate in (6). TS-JEPA scores $\mathbf{e}(t) = z_{\text{tgt}}(t) - \hat{z}_{\text{tgt}}(t)$ via $\mathcal{S}_{\text{TS}}(t) = g_D(\mathbf{e}(t))$. In 1D Soliton-Reynolds JEPA, we combine the joint MD-dimensional whitened gate $g_{MD}(\tilde{\mathbf{e}}(t))$ with the peak sub-horizon gate $\max_{1\le m\le M} g_D(\tilde{\mathbf{e}}_m(t))$ into $\mathcal{S}_{\text{env}}(t)$, modulated by a one-sided median absolute deviation (MAD)-standardized soliton concentration gain:

$$\mathcal{S}_{\text{env}}(t) = \sqrt{\frac{1}{2}\Big(g_{MD}\big(\tilde{\mathbf{e}}(t)\big)^2 + \max_{1\le m\le M} g_D\big(\tilde{\mathbf{e}}_m(t)\big)^2\Big)}, \tag{4}$$

$$\mathcal{S}_{\text{Sol-Rey}}(t) = \mathcal{S}_{\text{env}}(t)\Big(1 + \alpha_{\text{sol}} \max\big(0, \tanh(0.5\, z_{\text{LT}}(t))\big)\Big), \tag{5}$$

where $\alpha_{\text{sol}} = 0.12$ and $z_{\text{LT}}(t) = (R_{\text{LT}}(\mathbf{E}'(t)) - m_{\text{LT}})/(1.4826\,\text{MAD}_{\text{LT}})$ standardizes R_{LT} against its training median m_{LT} and MAD MAD_{LT} ($\alpha_{\text{sol}} = 0$ recovers $\mathcal{S}_{\text{env}}(t)$). *Three-Part Operational Causality Framework:* Online evaluation enforces three criteria (Fig. 2): (a) *Data causality ($\mathcal{F}_\tau$-measurability):* At decision step $\tau = t + S - 1$, encoders observe only $\mathbf{X}[t - C : \tau, :]$. (b) *Score-emission latency ($S - 1 = 63$ steps) vs. fault-detection delay (Δt):* Because z_{tgt} summarizes $\mathbf{X}[t : t + S, :]$, score s_t is emitted at $\tau = t + S - 1$ (Fig. 2, Row 3), creating a structural lag of $S - 1 = 63$ steps (0.98 s on 64 Hz Daphnet, 10.5 min on 0.1 Hz SMAP/MSL/GHL), distinct from empirical fault-detection delay $\Delta t = t_{\text{alarm}} - t^*$; shortening to $S \in \{1, 4\}$ eliminates emission lag (Section VII-D). (c) *Strictly causal trailing vs. retrospective buffered attribution: Causal trailing attribution* assigns s_t only to $\tau = t + S - 1$, whereas *buffered smear attribution* averages s_t over $[t, t + S - 1]$ (Table III(a),(b)).

D. Shared Architectural Components and Thresholding

1) Coordinate Saliency Gating and Scale Equivariance: To amplify sparse departures while mitigating channel drowning, we define $g_d : \mathbb{R}^d \to \mathbb{R}_{\ge 0}$ for $\mathbf{v} \in \mathbb{R}^d$:

$$g_d(\mathbf{v}) = (1-\alpha)\|\mathbf{v}\|_2 + \alpha\sqrt{d\sum_{i=1}^{d} w_i v_i^2}, \qquad w_i = \frac{\exp\big(|v_i|/(\tau_g\sigma_{\mathbf{v}})\big)}{\sum_{j=1}^{d}\exp\big(|v_j|/(\tau_g\sigma_{\mathbf{v}})\big)}, \quad \sigma_{\mathbf{v}} = \frac{\|\mathbf{v}\|_2}{\sqrt{d}} + \epsilon, \tag{6}$$

with $\alpha = 0.5$, $\tau_g = 0.5$, and $\epsilon = 10^{-6}$, satisfying approximate degree-1 homogeneity $g_d(c\mathbf{v}) \approx c\, g_d(\mathbf{v})$ for $\|\mathbf{v}\|_2 \gg \epsilon$. Input channels use uniform weights (Section VI-B).

2) Multiscale Frequency Front-End: For $K > 1$, a causal depthwise 1D convolutional stem filters each channel prior to 1×1 mixing ($K \le 1$ uses an identity bypass). Channels are split across $J = 4$ smooth dyadic shells on a 256-point rFFT grid normalized into an exact amplitude partition of unity ($\sum_{j=1}^{J} \Psi_j[\omega_m] \equiv 1$ within 1.2×10^{-7}), processed by causal 1D convolutions (kernels $\{15, 9, 5, 3\}$), and recombined via a softmax router, acting as a multiscale front-end rather than an orthogonal wavelet bank (Table II(b)).

3) Extreme-Value Threshold Calibration (SPOT): To convert scalar scores $\mathcal{S}(t)$ into binary alarms $\hat{y}_t = \mathbb{I}(\mathcal{S}(t) > \tau_{\text{SPOT}})$ from nominal calibration data alone, SPOT [8] fits exceedances $Y = \mathcal{S}(t) - u > 0$ above the 98th calibration percentile $u = p_{98}$ to a Generalized Pareto Distribution (GPD) [38] with shape ξ and scale σ via Grimshaw maximum-likelihood estimation [49] with bounded L-BFGS-B fallback ($\xi \in [-0.9, 10]$). Inverting the GPD tail at target risk $q = 10^{-3}$ yields:

$$\tau_{\text{SPOT}} = \begin{cases} u + \frac{\sigma}{\xi}\left[\left(\frac{qN_{\text{cal}}}{N_u}\right)^{-\xi} - 1\right], & |\xi| \ge 10^{-5}, \\ u - \sigma \ln\left(\frac{qN_{\text{cal}}}{N_u}\right), & |\xi| < 10^{-5}, \end{cases} \tag{7}$$

where N_{cal} and N_u count post-attribution calibration timestamps.

4) 224-Bit Multi-Perspective Circuit-Breaker Panel: When $MD = 128$ whitened sub-horizon residuals $\tilde{e}_{m,c}(t)$ ($m \in \{1, \ldots, 4\}, c \in \{1, \ldots, 32\}$) are collapsed into a single 1D score $\mathcal{S}(t)$ before calibration, one or two coordinates experiencing mild out-of-sample variance shift between training and calibration (*hijacker dials*) dominate $u = p_{98}$ and push τ_{SPOT} to the extreme tail tip, firing only on isolated peak steps inside an anomaly. To eliminate scalar score collapse and GPD tail extrapolation without test labels, the *224-bit multi-perspective circuit-breaker panel* expands each coordinate $c \in \{1, \ldots, D\}$ into seven fluid-dynamic perspectives ($7D = 224$ dials): (i) four sub-horizon flow-position magnitudes $|\tilde{e}_{m,c}(t)|$ (128 dials); (ii) Reynolds

horizon-mean transport $|\bar{e}_c(t)| = |M^{-1}\sum_{m=1}^{M}\tilde{e}_{m,c}(t)|$ (32 dials); (iii) Reynolds turbulent-fluctuation amplitude $\sigma'_c(t) = \sqrt{M^{-1}\sum_{m=1}^{M}(\tilde{e}_{m,c}(t)-\bar{e}_c(t))^2}$ (32 dials); and (iv) convective right-edge acceleration $|\tilde{e}_{M,c}(t)-\tilde{e}_{1,c}(t)|$ across $[48,64]$ vs. $[0,16]$ (32 dials). Nominal-only calibration applies two rules:

1) *Calibration-stabilized whitening and hijack guards:* Any coordinate whose calibration RMS exceeds 1.10 is blended toward its calibration-normalized unwhitened residual via weight $\lambda_{m,c} = \exp\big(-1.5\max(0, \mathrm{RMS}_{\mathrm{cal}}(m,c) - 1.10)\big)$. Each dial $j \in \{1,\dots,224\}$ is standardized by its calibration median and MAD into $z_j(t)$ with binary tripwire $b_j(t) = \mathbb{I}(z_j(t) > \tau_j)$, where $\tau_j = q_{1-\alpha}(z_{j,\mathrm{cal}})\,\gamma_j^{\mathrm{cal}}\,\gamma_j^{\mathrm{tr}\to\mathrm{cal}}$ ($\alpha = 0.002$) scales the calibration quantile by intra-calibration drift γ_j^{cal} and train-to-calibration tail growth $\gamma_j^{\mathrm{tr}\to\mathrm{cal}}$.
2) *Boolean quorum and fast right-edge onset relays:* Let $N_{\mathrm{trip}}(t) = \sum_{j=1}^{224} b_j(t)$ count tripped switches. *224-Bit Strict* alerts when $N_{\mathrm{trip}}(t) \geq K_{\mathrm{strict}}$ (99.3rd calibration percentile). *224-Bit Breaker* alerts when $N_{\mathrm{trip}}(t) \geq K_{\mathrm{bal}}$ (98.5th percentile) *or* any of 32 fast right-edge onset relays trips: $b_{M,c}(t) \wedge \big(b_{\mathrm{accel},c}(t) \vee b_{\mathrm{fluc},c}(t) \vee b_{M-1,c}(t)\big)$. Under buffered attribution, each dial votes only over the prefix steps $[t, t+S_m-1]$ (or $[t+S_{M-1}, t+S_M-1]$ for right-edge acceleration) that it physically observes.

V. Experimental Protocol and Evaluation Metrics

A. Experimental Setup and Hyperparameters

All architectures use an identical streaming pipeline with context window $C = 256$ and target horizon $S = 64$ ($W = C + S = 320$), trained at stride 16 and evaluated at stride 1. Context and target representations ($D = 32$) are extracted by the multiscale frequency front-end followed by a 6-block causal dilated TCN (64 filters, kernel size 5, dilations $\{1,2,4,8,16,32\}$). Multi-statistic temporal pooling (terminal, mean, max, standard deviation, Log-Ricci energy, and first-order total variation) projects features through a linear bottleneck before LayerNorm and dropout ($p = 0.20$). Models are trained for 20 epochs on nominal sequences using AdamW [50] (learning rate $10^{-3} \to 10^{-5}$ via cosine annealing, weight decay 10^{-4}, batch size 32), with identical training budgets and zero per-dataset hyperparameter tuning across families.

a) Held-Out Calibration Protocol and Boundary Isolation: To evaluate out-of-sample threshold calibration without circularity, each stream's raw nominal sequence $\mathbf{X}_{\mathrm{nom}} \in \mathbb{R}^{T_{\mathrm{nom}}\times K}$ is partitioned chronologically (Fig. 2, Row 1) into a *training block* $\mathbf{X}_{\mathrm{tr}} = \mathbf{X}_{\mathrm{nom}}[0:N_{\mathrm{tr}})$ ($N_{\mathrm{tr}} = \lfloor 0.65\,T_{\mathrm{nom}}\rfloor$) and a disjoint *held-out nominal block* $\mathbf{X}_{\mathrm{val}} = \mathbf{X}_{\mathrm{nom}}[N_{\mathrm{tr}} : T_{\mathrm{nom}})$. Standardization parameters and neural weights are fit on $\mathbf{X}_{\mathrm{tr}}$ windows alone, guaranteeing zero raw-observation or window overlap across N_{tr}.

Within $\mathbf{X}_{\mathrm{val}}$ ($N_{\mathrm{val}} = T_{\mathrm{nom}} - N_{\mathrm{tr}}$), sliding windows of length $W = 320$ are constructed strictly inside $\mathbf{X}_{\mathrm{val}}$ (Fig. 2, Row 2). Under causal trailing attribution, the first $W-1 = 319$ observations provide context warm-up, yielding $M_{\mathrm{val}} = N_{\mathrm{val}} - W + 1$ valid score timestamps $v[0 : M_{\mathrm{val}})$. Splitting v at

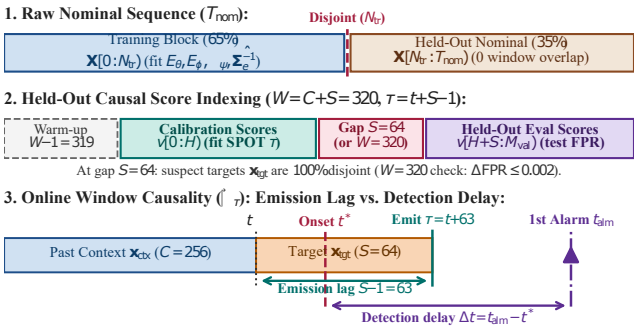


Fig. 2. Held-out evaluation protocol (C=256, S=64, W=320): (Row 1) chronological train/validation split, (Row 2) boundary-isolated calibration and evaluation blocks, and (Row 3) score-emission lag versus fault-detection delay.

midpoint $H = \lfloor M_{\mathrm{val}}/2\rfloor$ defines the *nominal calibration block* $v_{\mathrm{cal}} = v[0 : H)$ (used to fit the SPOT threshold at $u = p_{98}, q = 10^{-3}$) and the *held-out evaluation block* $v_{\mathrm{hold}} = v[H+S : M_{\mathrm{val}})$, excluding $S = 64$ intermediate windows $v[H : H+S)$. Because suspect target windows have length $S = 64$, the last calibration target and first evaluation target are 100% disjoint. The first $C = 256$ evaluation steps share passive historical context with calibration observations, mirroring online streaming where past nominal history forms context for current predictions. A full-window isolation check ($G = W = 320$, excluding $v[H : H+320)$ across the seven streams with $M_{\mathrm{val}}/2 > 420$) confirms that shared passive context does not bias calibration: stream-macro causal FPR changes by less than 0.25 percentage points for both JEPAs (0.33% to 0.36% for Soliton-Reynolds JEPA and 3.02% to 3.24% for TS-JEPA), remaining an order of magnitude below TimesNet and TranAD (9.7%–14.0%). All primary tables use fixed standard SPOT thresholds during testing.

b) Independent Statistical Units, Multi-Seed Replication, and Equivalence Bounds: Principal models on the 19 non-GHL streams are evaluated across five random seeds ($\{42, 123, 456, 7, 2024\}$); the 14 GHL streams ($> 10^6$ observations each) and ablation controls use three seeds ($\{7, 42, 2024\}$). To prevent pseudoreplication, seeds are averaged within each stream first ($\mathrm{SD}_{\mathrm{seed}}$), stream-level paired tests treat the telemetry stream ($n = 33$ primary or $n = 19$ non-GHL) as the unit of analysis, and domain-level tests treat the dataset family ($N = 7$) as the independent cluster:

1) *Paired Wilcoxon signed-rank test and false-discovery-rate adjustment:* The Wilcoxon signed-rank test evaluates whether the median paired difference across streams is zero without Gaussian assumptions. We adjust p-values across multiple comparisons via the Benjamini–Hochberg false-discovery-rate (FDR) procedure [51] ($q_{\mathrm{FDR}} = 0.05$), report 95% stream-resampled bootstrap confidence intervals (CIs; 2,000 resamples), and cross-check at the dataset-cluster level ($N = 7$) via exact sign-flip permutation tests and zero-tuning LODO sensitivity.
2) *Two One-Sided Tests (TOST) for fixed-backbone regularizer equivalence:* We use TOST [15] to test whether the multiplicative soliton and manifold gain in (5) alters aggregate Point-F1 relative to the unmodulated multi-horizon whitened envelope $\mathcal{S}_{\mathrm{env}}(t)$ on the identical trained backbone

($n = 19$ streams). TOST tests $H_{01} : \mu_1 - \mu_2 \leq -\Delta$ (p_{lower}) and $H_{02} : \mu_1 - \mu_2 \geq +\Delta$ (p_{upper}), establishing equivalence when $p_{\text{TOST}} = \max(p_{\text{lower}}, p_{\text{upper}}) < 0.05$ across $\Delta \in \{\pm 0.015, \pm 0.020, \pm 0.030\}$ (Table II(a)).

3) *Distributional tail and drift diagnostics (γ_2 and D_{KS}):* On nominal windows we measure: (i) *excess kurtosis* $\gamma_2 = \mathbb{E}[(s-\mu)^4]/\sigma^4 - 3$, distinguishing heavy-tailed *leptokurtic* ($\gamma_2 > 0$) from light-tailed *platykurtic* ($\gamma_2 < 0$) residuals; and (ii) the two-sample *Kolmogorov–Smirnov (KS) drift statistic* $D_{\text{KS}} = \sup_x |F_{\text{cal}}(x) - F_{\text{hold}}(x)| \in [0,1]$, quantifying distributional shift between calibration (F_{cal}) and held-out nominal (F_{hold}) blocks.

B. Baseline Fairness Checks, Cohort Structure, and Reproducibility

Two fairness audits verify that reconstructive baseline results under the unified 20-epoch, $W = 320$ pipeline are not artifacts of window length or implementation: (i) evaluating TranAD ($W = 10$) and TimesNet ($W = 96$) at their native published window sizes across five streams raises Point-F1 to 0.164 and 0.218, but observation residuals remain severely heavy-tailed ($\gamma_2 > 98$ vs. 3.1 for TS-JEPA) with high nominal FPR (28.4%–44.1%); and (ii) reconstructors replicate their published point-adjusted F1 on SMD and MSL, while Table III(d) provides a secondary 130-stream, 11-dataset single-seed cross-check against NCAD, DCdetector, and Anomaly Transformer.

Experiments span five cohorts: *Cohort 1* (33 primary streams: 19 non-GHL at 5 seeds and 14 GHL at 3 seeds, Tables III(a)–(c) and V(a)); *Cohort 2* (19-stream, 12-arm matched-backbone factorial and operating-curve suite at 5 seeds, Table II(a),(c) and Fig. 4(b)); *Cohort 3* (6-stream single-channel quiet-fault and hybrid-monitor suite at 3 seeds, Fig. 3); *Cohort 4* (5-stream horizon-latency sweep across $S \in \{1, 4, 16, 32, 64\}$ and dyadic front-end ablation at 3 seeds, Fig. 4(a) and Table II(b)); and *Cohort 5* (141-stream, 3-seed SPOT benchmark across 12 multivariate datasets with validation early stopping plus the 130-stream single-seed baseline screen, Table III(d)). At $K = 25$ channels, online parameter counts are matched: 272k for TS-JEPA (540k with EMA target), 312k (619k total) for Soliton-Reynolds JEPA, 304k–307k for TCN observation forecasters, 649k for TimesNet, and 206k for TranAD. Non-finite entries are zero-imputed ($\boldsymbol{\sigma} \geq 10^{-2}$), and causal and buffered scores receive separately fitted thresholds τ_{SPOT} on v_{cal} (7). Runs executed on PyTorch 2.4 / CUDA 12.4 (RTX 5090, 0.60–0.88 ms/window, 188 MB memory); code is at https://github.com/quaid281/NCAD.

C. Multi-Tier Statistical Evaluation Metrics

To guard against point-adjustment inflation, channel-averaging bias, and threshold gaming [16], [40], we evaluate models across three tiers. First, *primary calibration and accuracy outcomes* comprise realized held-out nominal FPR (FPR_{hold}), out-of-sample calibration error $|\text{FPR}_{\text{hold}} - q|$ relative to design risk $q = 10^{-3}$, unadjusted Point-F1, and threshold-free PR-AUC. Second, *operating characteristics* evaluate event recall across deployable calibration quantiles (q^*, Fig. 4(b)) and across post-hoc matched realized test-FPR budgets ($f^* \in \{0.01, 0.05\}$, Table II(c)). Third, *secondary operational endpoints* report unweighted Dataset-Macro Point-F1, event recall (fraction of contiguous anomaly segments with ≥ 1 alarm), mean detection delay Δt over detected events, and the constant-alarm baseline $\text{F1}_{\text{all-pos}} = 2\pi/(1+\pi)$ at prevalence π.

VI. Ablation Studies and Analysis

A. Ablation 1: Multi-Horizon Whitened Envelope vs. Soliton Concentration Modulation

To separate the multi-horizon representation from post-hoc scoring multipliers on the identical trained network, Table II(a) and Fig. 3(a) compare single-horizon TS-JEPA against the unmodulated multi-horizon envelope control (α_{sol}=0, outputting $\mathcal{S}_{\text{env}}(t)$ directly) and a sweep across $\alpha_{\text{sol}} \in [0.03, 0.24]$ on all 19 non-GHL streams. Replacing single-horizon TS-JEPA with the unmodulated multi-horizon whitened envelope $\mathcal{S}_{\text{env}}(t)$ drives the core improvements in false-alarm control and low-FPR sensitivity (Table II(a),(c)): causal held-out FPR drops four-fold from 2.4% to 0.6%, calibration drift decreases, and causal event recall at a 1% false-alarm budget rises from 0.395 to 0.578 without loss of buffered PR-AUC. Across $\alpha_{\text{sol}} \in [0, 0.24]$, adding the post-hoc Lieb–Thirring soliton concentration gain yields a modest buffered FPR reduction (8.3% to 7.8% at $\alpha_{\text{sol}} = 0.12$, Wilcoxon $p = 0.026$) and nudges causal event recall from 0.439 to 0.478, while buffered Point-F1 and PR-AUC satisfy TOST equivalence within $\Delta = \pm 0.020$ ($p_{\text{TOST}} = 2.7\times10^{-5}$, Table II(a)). Auxiliary spectral regularizers during training similarly leave downstream recall and FPR flat; the primary gains come from multi-depth TCN tapping, barycenter centering, and multi-horizon Mahalanobis whitening.

B. Ablation 2: Single-Channel Quiet Faults, Input Gating, and Hybrid Monitoring

When one sensor out of K freezes at a nominal value while the remaining $K-1$ channels fluctuate normally, the frozen channel perturbs the pooled pre-LayerNorm vector by only $O(1/K)$, and LayerNorm rescales the embedding onto $\sqrt{D}\mathbb{S}^{D-1}$ using the remaining active channels. Pre-weighting input channels by context variance lowers macro Point-F1 across four multivariate streams from 0.118 to 0.077, so the deployed pipeline retains uniform input encoding. Across six single-channel fault classes injected into six multivariate streams (Fig. 3(b)), uniform TS-JEPA detects 27.8% of abrupt $+5\sigma$ spikes, but on persistent quiet faults (flatlines, stuck sensors, and zero dropouts) event recall falls to 19.4% and point recall drops below 3%. Simple observation-space monitors catch these complementary failures: a maximum channel deviation monitor and a rolling variance-collapse monitor (tracking how far each channel's 32-step trailing standard deviation drops below its nominal median) each recover 25%–39% of quiet-fault events at 14%–15% FPR. Taking the logical union of TS-JEPA alarms and the rolling variance-collapse monitor lifts event recall to 41.7%–50.0% across all six fault classes at 17.0% nominal FPR.

TABLE II
ARCHITECTURAL ABLATION AND MECHANISM-DISCRIMINATION STUDIES (n=19 STREAMS, 5 SEEDS; EVENT RECALL (ER)).

(a) Fixed-Backbone Soliton Control (n=19)

Model	Pt-F1	PR	FPR	ER	ΔF1 [90%]	p_{TOST}
TS-JEPA (Unconstrained)	0.206 ± 0.025	0.331	0.093	0.477	Baseline	–
Soliton-Reynolds JEPA	0.148 ± 0.039	**0.331**	**0.078**	0.430	−0.001 [−.007, +.005]	$< 10^{-6*}$
$\alpha_{\text{sol}} = 0$ control	0.149 ± 0.038	0.330	0.083	0.462	–	$(< 10^{-4\dagger})$
TimesNet	0.197 ± 0.018	0.329	0.300	**0.583**	–	–
TranAD	0.130 ± 0.004	0.287	0.307	0.490	–	–

* TOST equivalence vs. α_{sol}=0 control at $\Delta = \pm 0.030$ ($p = 1.1 \times 10^{-7}$); † at $\Delta = \pm 0.020$ ($p = 2.7 \times 10^{-5}$; Wilcoxon p=0.472). Matched-FPR ER (f^*=0.01): causal 0.562 vs. 0.578; at f^*=0.05: 0.698 vs. 0.711.

(b) Dyadic Front-End (n=5)

Stream (F1 / PR)	Std. Conv	Dyadic	ΔF1 / ΔPR
SMAP P-3	**0.301** / 0.280	0.115 / **0.296**	−0.186 / +0.017
MSL M-1	0.014 / **0.491**	**0.016** / 0.449	+0.002 / −0.043
SMD machine-1-2	0.000 / 0.028	**0.068** / **0.221**	+0.068 / +0.193
Daphnet S01R01E1	**0.179** / **0.103**	0.130 / 0.066	−0.048 / −0.037
GECCO water-quality	**0.097** / 0.161	0.062 / **0.282**	−0.035 / +0.121
Macro Mean	**0.118** / 0.213	0.078 / **0.263**	−0.040 / +0.050

(c) 12-Arm Matched-Backbone Factorial Comparison (n=19 Streams, 5 Seeds)

Architecture Arm	Tail / Drift γ_2	Tail / Drift D_{KS}	Held-Out FPR Caus.	Held-Out FPR Buf.	Point-F1 Caus.	Point-F1 Buf.	PR-AUC Caus.	PR-AUC Buf.	Causal Event Recall SPOT	Causal Event Recall q^*=0.05	Causal Event Recall f^*=0.01	Causal Event Recall f^*=0.05
TS-JEPA (Unconstrained)	1.33	0.214	0.024	0.093	0.161	0.206	0.278	**0.331**	0.487	0.730	0.395	0.584
Soliton-Reynolds JEPA	2.30	0.179	**0.006**	0.078	0.050	0.148	0.259	**0.331**	0.478	0.752	**0.562**	**0.698**
Soliton-Reynolds (α_{sol}=0)	2.19	0.183	**0.006**	0.083	0.055	0.149	0.261	0.330	0.439	0.755	**0.578**	**0.711**
JEPA Context Autoencoder	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	0.431	0.573
JEPA Hard-Target Copy	1.16	0.167	0.009	0.098	0.146	**0.235**	0.273	0.328	0.503	0.816	0.398	0.600
TS-JEPA (ℓ_2 Scoring)	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
PCA Linear Forecaster (LN)	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	0.624
TCN Obs. Forecaster (LN)	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
Isolation Forest	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
TCN Obs. Forecaster	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
TCN Obs. Autoencoder	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
TimesNet	5.54	0.438	0.243	0.300	**0.209**	0.197	0.281	0.329	**0.588**	0.618	0.217	0.407
TranAD	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

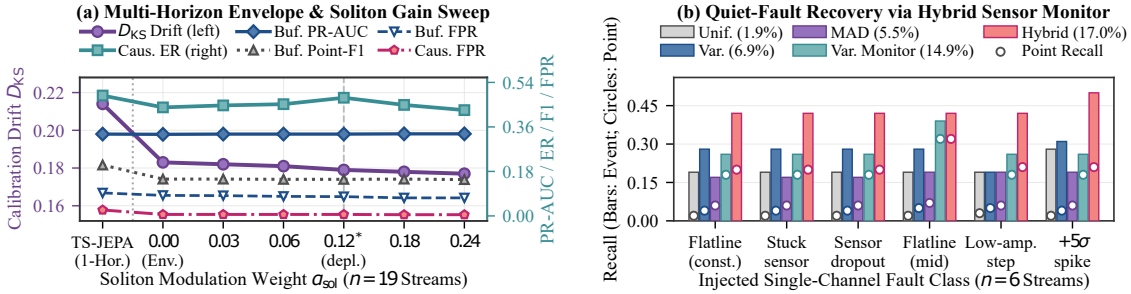


Fig. 3. Multi-horizon soliton modulation sweep and quiet-fault recovery. **(a)** Calibration-to-holdout drift (D_{KS}, left axis) and buffered PR-AUC, causal ER, buffered Point-F1, buffered FPR, and causal FPR (right axis) across single-horizon TS-JEPA and Soliton-Reynolds JEPA modulation weights $\alpha_{\text{sol}} \in [0, 0.24]$ (n=19 streams, five seeds). **(b)** Injected single-channel ER (bars) and point recall (circles) across uniform TS-JEPA (Unif.), context-variance (Var.) and median absolute deviation (MAD) input gates, the standalone rolling variance-collapse monitor (Var. Monitor), and the hybrid union detector (Hybrid; nominal FPR in parentheses, n=6 streams, three seeds).

C. Ablation 3: Multiscale Dyadic-Shell Front-End

Table II(b) trains TS-JEPA with the identical 3-layer causal TCN encoder across five streams, comparing the 4-shell Fourier filterbank against a standard linear convolutional stem. Enabling the dyadic front-end raises macro PR-AUC from 0.213 to 0.263 (led by SMD and GECCO) and lowers validation residual kurtosis from 1.41 to 1.07, while macro Point-F1 decreases from 0.118 to 0.078 due to a more conservative SPOT threshold on SMAP P-3. Frequency-shell splitting thus improves score ranking and tail conditioning rather than uniformly raising thresholded F1.

D. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Comparing the 12 model arms across all 19 non-GHL streams under identical splits, training budgets, and matched parameter counts in Table II(c) isolates five architectural factors:

1) *Forecasting vs. reconstructing raw observations (TCN Obs. Forecaster vs. TCN Obs. Autoencoder):* Pairing the same causal dilated TCN encoder with a multilayer perceptron decoder to either forecast the 64-step target window or reconstruct the 320-step window in raw sensor space leaves tail kurtosis ($\gamma_2 \approx 6.2$), calibration drift ($D_{\text{KS}} \approx 0.45$), and causal FPR ($\approx 25\%$) nearly identical.
2) *LayerNorm bottleneck and saliency gating in observation space (TCN Obs. Forecaster (LN)):* Adding a $D = 32$ LayerNorm bottleneck, target-window LayerNorm, and coordinate saliency gating to the observation forecaster compresses tail kurtosis six-fold ($\gamma_2 = 0.91$) and halves causal FPR to 11.7%, but targeting raw sensor waveforms leaves calibration drift high ($D_{\text{KS}} = 0.376$) and drops causal Point-F1 to 0.084.
3) *Learned latent target projection and multi-horizon whitening (TS-JEPA and Soliton-Reynolds JEPA):* Moving the prediction target into the learned $D = 32$ latent space cuts calibration drift nearly in half ($D_{\text{KS}} = 0.214$ in TS-JEPA and 0.179 in Soliton-Reynolds JEPA, paired Wilcoxon $p \leq 0.001$), lowering causal FPR to 2.4% and 0.6% while lifting buffered PR-AUC from 0.257 to 0.331.
4) *Nonlinear TCN encoding vs. linear PCA and tree isolation (PCA Linear Forecaster (LN) and Isolation Forest):* Replacing the nonlinear TCN with 32 principal components and ridge regression bounds causal FPR to 2.1% through Layer-

Norm, but causal Point-F1 collapses to 0.017 (and Isolation Forest reaches only 0.149 causal event recall), confirming the need for learned nonlinear temporal features.

5) *Future prediction vs. context autoencoding, EMA stability, and ℓ_2 scoring (JEPA Context Autoencoder, JEPA Hard-Target Copy, and TS-JEPA (ℓ_2 Scoring)):* Context autoencoding in latent space keeps causal FPR low (1.7%) but drops causal Point-F1 from 0.161 to 0.112 and PR-AUC from 0.278 to 0.235. Direct weight copying ($\zeta = 0$) or replacing the saliency gate with a Euclidean norm still holds causal FPR below 2% (0.9% and 1.8%), confirming that false-alarm reduction stems from the normalized latent representation.

VII. Results and Discussion

A. Benchmark Performance on Continuous Telemetry

Table III reports all 33 continuous telemetry streams under the chronological held-out SPOT protocol ($u = p_{98}, q = 10^{-3}$): Panel (a) uses retrospective buffered attribution, Panel (b) strictly causal streaming, and Panel (c) per-dataset breakdowns including anomaly event counts (E) and both causal and buffered held-out nominal FPRs. Evaluating realized FPR and out-of-sample calibration error $|\text{FPR}_{\text{hold}} - q|$ as primary outcomes shows a large relative reduction in false alarms and calibration error, while Soliton-Reynolds JEPA achieves the lowest held-out FPR across both attribution regimes (0.6% causal, 7.8% buffered), matches TS-JEPA on buffered PR-AUC (0.331), and leads in causal matched-FPR event recall (0.562 at $f^* = 0.01$).

Lower Realized FPR and Calibration Error on Held-Out Telemetry: On the 19 non-GHL streams (Table III(a),(b)), latent scoring cuts held-out nominal FPR and calibration error $|\text{FPR}_{\text{hold}} - q|$ three-fold under buffered attribution (7.8% for Soliton-Reynolds JEPA and 9.3% for TS-JEPA vs. $\approx 30\%$ for TimesNet and TranAD, $p \leq 0.045$). Under strictly causal streaming, the false-alarm reduction widens to ten- to forty-fold: 0.6% for Soliton-Reynolds JEPA and 2.4% for TS-JEPA, compared to 22.2%–24.3% for the reconstructors ($p = 0.010$). This separation holds across all 33 streams (0.38% and 1.51% causal FPR vs. 14.2%–16.1% for reconstructors, Table V(a)), across all $N = 7$ dataset families ($p = 0.016$), and under zero-tuning LODO evaluation (0.3%–2.3% causal FPR for JEPAs vs. 12.1%–18.4% for reconstructors). In Table III(c), Soliton-Reynolds JEPA achieves the lowest causal held-out FPR on every multi-stream family (0.1%–0.8% vs. 5.6%–36.0% for baselines).

Multi-Horizon Ranking Gains vs. Unadjusted Point-F1 Trade-offs and 224-Bit Circuit-Breaker Resolution: Under buffered attribution on the 19 non-GHL streams (Table III(a)), TS-JEPA and Soliton-Reynolds JEPA tie for the highest PR-AUC (0.331 vs. 0.329 for TimesNet and 0.287 for TranAD). Under scalar 1D SPOT, Soliton-Reynolds JEPA trades lower buffered Point-F1 than TS-JEPA (0.148 vs. 0.206) for tighter false-alarm control, outperforming TranAD (0.130 F1) at one-quarter the FPR (7.8% vs. 30.7%). Across all 141 qualifying multivariate streams in 12 datasets at three seeds with validation early stopping (Table III(d)), 3-horizon Speculative-JEPA and Speculative-Alien-JEPA (adding a coreset and residual null-space bank) surpass single-horizon TS-JEPA both causally (0.184 vs. 0.165 PR-AUC, 95% CI $[+0.006, +0.034]$; 0.077 vs. 0.067 F1) and under buffered attribution (0.253–0.256 vs. 0.213 PR-AUC, $p \leq 0.0042$, and 0.567 vs. 0.459 event recall, $p = 1.6\times10^{-4}$). Replacing 1D scalar collapse and SPOT extrapolation with the 224-bit multi-perspective circuit-breaker panel (Table III(d)) resolves the Point-F1 trade-off across all 141 streams: causally, *Speculative-224Bit-Strict* and *Speculative-224Bit-Breaker* lift PR-AUC to 0.197 ($p = 0.014$ vs. TS-JEPA), nearly double causal Point-F1 from 0.067–0.077 to 0.125–0.139 (97 wins vs. 36 losses over TS-JEPA, $p = 8.3\times10^{-11}$; 114 wins vs. 18 losses over 1D Speculative-Alien-JEPA, $p = 1.2\times10^{-16}$), raise causal event recall from 48.0%–50.7% to 64.8%–72.7%, and cut mean detection delay from 115.4–125.4 to 55.3–78.0 steps. Under buffered attribution across the 141 streams, the 224-bit panel raises PR-AUC to 0.261 ($p = 0.0031$), lifts Point-F1 to 0.177–0.180 ($p \leq 5.7\times10^{-4}$ vs. TS-JEPA and $p \leq 2.6\times10^{-4}$ vs. 1D Speculative-Alien-JEPA), raises event recall to 66.0%–68.5%, and lowers buffered held-out FPR below 1D Speculative-Alien-JEPA (16.5% vs. 18.0%) while cutting delay to 46.6–54.0 steps. On GHL ($\pi = 0.013$, Table III(c)), all JEPAs and the linear autoregressive (AR) forecaster fire near-silently (0.1%–0.4% FPR, 0.000 F1), showing that low FPR without recall does not indicate detection.

Domain and Operational Applicability Boundaries: Synthesizing performance across fault classes and sampling rates defines four operational regimes:

1) *Favorable operating regime:* Multi-channel spacecraft and wearable kinematic arrays ($K \geq 9$, SMAP, MSL, Daphnet) with persistent localized transients, where suppressing nuisance false alarms under automated calibration is the primary operational requirement.
2) *Latency-constrained telemetry:* Systems where a 63-step emission lag is prohibitive for immediate tripwire response (such as hourly Metro where 63 steps $= 63$ h), requiring short horizons $S \in \{1, 4\}$ (Section VII-D).
3) *Quiet single-channel faults:* Inactive sensor modes (flatlines, dropouts) that LayerNorm embeddings absorb as nominal background, requiring the hybrid union detector (combining TS-JEPA with a rolling variance-collapse monitor) when single-channel coverage is mandatory (Section VI-B).
4) *Continuous non-stationary process ramps:* Industrial heating loops and traffic drifts (GHL, Metro; 0.000–0.022 F1), which defeat both static and rolling thresholds.

B. Component-Level Analysis of Soliton-Reynolds JEPA

Tables II(a),(c) and III(a),(b) isolate how each component of Soliton-Reynolds JEPA modifies performance relative to TS-JEPA. First, *multi-depth TCN tapping and barycenter centering* remove common-mode baseline shift, lowering calibration drift (D_{KS} from 0.214 to 0.179) and cutting the inter-block drift share of excess causal FPR four-fold (71.3% to 18.7%, Table V(b)). Second, *nested multi-horizon rollout and Mahalanobis whitening* ($\mathcal{S}_{\text{env}}(t)$ at $\alpha_{\text{sol}} = 0$) capture sub-

TABLE III
HELD-OUT CALIBRATION BENCHMARK UNDER SPOT ($u{=}p_{98}, q{=}10^{-3}$; BEST IN **BOLD**, SECOND UNDERLINED).

	(a) Buffered Attribution ($n{=}19$ Non-GHL)					(b) Causal Attribution ($n{=}19$ Non-GHL)				GHL ($n{=}14$)		Macro
Architecture	**FPR ↓**	**Pt-F1 ↑**	**PR-AUC ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**DS-F1 ↑**
Latent-space predictive architectures (JEPAs)												
TS-JEPA (unconstrained)	0.093	**0.206**	**0.331**	0.477	110.3	0.024	0.161	0.487	124.0	0.004	0.000	**0.115**
Soliton-Reynolds JEPA	0.078	0.148	**0.331**	0.430	127.1	**0.006**	0.050	0.478	154.0	**0.002**	0.000	0.073
Soliton-Reynolds ($\alpha_{\text{sol}}{=}0$)	0.083	0.149	0.330	0.462	113.3	**0.006**	0.055	0.439	138.4	**0.002**	0.000	0.074
Observation-space reconstructive baselines												
TimesNet	0.300	0.197	0.329	**0.583**	**48.5**	0.243	**0.209**	**0.588**	**62.2**	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.140	0.533	241.0	0.046	0.020	0.069
Non-neural linear baseline												
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Dataset Summary					Buffered Point-F1 ↑				Held-Out FPR (Causal / Buffered) ↓			
Dataset	n	K	E	π	$\text{F1}_{\text{all-pos}}$	**TS-JEPA**	**Sol.-Rey.**	**TimesNet**	**TranAD**	**TS-JEPA**	**Sol.-Rey.**	**TimesNet**	**TranAD**
Daphnet	6	9	72	0.148	0.256	0.145	0.143	**0.229**	0.144	.030/.091	**.008/.072**	.291/.313	.250/.298
GHL	14	16	26	0.013	0.026	0.000	0.000	**0.022**	0.020	.004/.004	**.001/.002**	.056/.061	.040/.046
Genesis	1	18	2	0.003	0.006	0.000	**0.012**	0.000	0.000	.004/.017	.003/.098	**.000/.000**	**.000/.000**
MSL	4	55	6	0.255	0.381	**0.161**	0.081	0.158	0.151	.054/**.095**	**.004**/.098	.161/.133	.132/.241
SMAP	6	25	7	0.150	0.236	**0.379**	0.267	0.289	0.167	.014/.135	**.006/.093**	.333/.549	.360/.513
Metro	1	5	25	0.001	0.003	0.000	**0.001**	**0.001**	0.000	.008/.014	.002/.007	**.000/.000**	**.000/.000**
SWAN	1	38	6615	0.279	0.437	**0.119**	0.008	0.000	0.000	**.000/.000**	.001/**.000**	**.000/.000**	**.000/.000**

Note: 95% stream-bootstrap CIs ($n{=}19$) for (a) Buffered FPR/F1: TS-JEPA [.048, .145]/[.113, .319], Sol.-Rey. [.032, .137]/[.081, .229], ($\alpha_{\text{sol}}{=}0$) [.032, .145]/[.084, .226], TimesNet [.162, .446]/[.098, .315], TranAD [.177, .445]/[.053, .232], Linear AR [.001, .011]/[.012, .104]; (b) Causal FPR/F1: TS-JEPA [.007, .046]/[.063, .287], Sol.-Rey. [.003, .011]/[.013, .097], ($\alpha_{\text{sol}}{=}0$) [.003, .012]/[.014, .109], TimesNet [.123, .375]/[.109, .326], TranAD [.119, .337]/[.058, .241]. Causal alert rate $\hat{p}_{\text{alert}}$/TPR/$\text{ER}_{\text{rand}}(\text{FPR}_{\text{hold}})$ ($\text{ER}_{\text{rand}}(\hat{p}_{\text{alert}})$): TS-JEPA .108/.166/.283 (.656), Sol.-Rey. .024/.040/.272 (.491), ($\alpha_{\text{sol}}{=}0$) .025/.045/.278 (.487), TimesNet .242/.340/.600 (.801), TranAD .213/.239/.601 (.801). **(d) Archive Checks & 224-Bit Circuit-Breaker Panel:** *130-Stream Screen (Pt-F1, macro/stream):* NCAD .058/.088; DCdetector .070/.097; Anomaly Trans. .091/.082; TimesNet .129/.151; TranAD .156/.152; TS-JEPA **.160/.186**. *19-Stream, 3-Seed Soliton-Reynolds 224-Bit Breaker (Causal/Buffered Pt-F1, PR-AUC, ER, Δt):* 1D SPOT .052/.137, .291/.370, .398/.377, 246.6/402.7; 224-Bit Strict .116/.190, **.312**/**.411**, .534/.545, 248.0/97.3; 224-Bit Breaker **.138**/**.228**, **.312**/**.411**, **.607**/**.631**, **105.9**/**71.8**. *141-Stream, 3-Seed Benchmark (12 Datasets; Causal/Buffered Pt-F1, PR-AUC [ER, Δt]):* TS-JEPA .067/.142, .165/.213 [.480/.459, 115.4/117.9]; Speculative-JEPA .071/.155, .183/.256 [.497/.565, 123.0/65.8]; Speculative-Alien-JEPA .077/.153, .184/.253 [.507/.567, 125.4/65.1]; Speculative-224Bit-Strict .125/.177, **.197**/**.261** [.648/.660, 78.0/54.0]; Speculative-224Bit-Breaker **.139**/**.180**, **.197**/**.261** [**.727**/**.685**, **55.3**/**46.6**].

horizon transients across $S_m \in \{16, 32, 48, 64\}$, reducing causal FPR from 2.4% to 0.6% and lifting causal event recall at a 1% FPR budget from 0.395 to 0.578. Third, *post-hoc Lieb–Thirring soliton modulation* ($\alpha_{\text{sol}} = 0.12$) trims buffered FPR slightly (8.3% to 7.8%, $p = 0.026$) and nudges causal event recall from 0.439 to 0.478, but leaves Point-F1 and PR-AUC statistically unchanged on the fixed backbone ($p_{\text{TOST}} < 10^{-4}$), confirming that the multi-horizon whitened representation drives the primary gains.

C. Operational Incident Evaluation and Causal-vs-Buffered Trade-offs

Comparing Panels (a) and (b) of Table III shows how zero-lookahead streaming shifts the operating threshold under 1D SPOT. Without 64-step window averaging, causal 128-dimensional whitened residuals have a heavier right tail ($\gamma_2 = 2.30$) than buffered scores ($\gamma_2 = 1.25$) or single-horizon TS-JEPA ($\gamma_2 = 1.33$). When 1D SPOT extrapolates that tail to $q = 10^{-3}$ in (7), one or two out-of-sample drifting coordinates (*hijacker dials*) place τ_{SPOT} high on the causal score distribution. Consequently, 1D Soliton-Reynolds JEPA alerts on only $\hat{p}_{\text{alert}} = 2.4\%$ of test steps (4.0% true-positive rate (TPR) at 0.6% held-out FPR vs. 10.8% for TS-JEPA and 24.2% for TimesNet), flagging only the sharpest peak steps inside each anomaly segment (0.050 causal Point-F1 despite 47.8% event recall). Against an analytical rate-matched i.i.d. Bernoulli random alerter $\text{ER}_{\text{rand}}(p) = E^{-1}\sum_{e=1}^{E}[1-(1-p)^{L_e}]$ (Table III Note), random firing at Soliton-Reynolds JEPA's held-out nominal rate ($p = 0.56\%$) catches only 27.2% of events versus 47.8% actual, whereas for TimesNet (24.3% FPR) random firing alone reaches 60.0%, exceeding TimesNet's actual 58.8%.

Replacing 1D scalar collapse on the 19 non-GHL streams with the *224-bit multi-perspective circuit-breaker panel* (Section IV-D4, Table III(d)) eliminates hijacker-dial threshold inflation: in strictly causal streaming, *224-Bit Strict* more than doubles Soliton-Reynolds JEPA's Point-F1 from 0.052 to 0.116 (16 wins / 1 loss, $p = 3.1{\times}10^{-5}$), lifts causal PR-AUC from 0.291 to 0.312, and raises event recall from 39.8% to 53.4% at 1.06% held-out FPR, while *224-Bit Breaker* (adding the 32 right-edge onset relays) reaches 0.138 causal Point-F1 (17 wins / 0 losses, $p = 1.5{\times}10^{-5}$), 60.7% event recall, and cuts detection delay from 246.6 to 105.9 steps (2.3× faster) at 1.67% FPR. Under buffered attribution with prefix-horizon routing, *224-Bit Strict* and *224-Bit Breaker* lift buffered PR-AUC from 0.370 to 0.411, Point-F1 from 0.137 to 0.190–0.228 ($p = 0.026$), and event recall from 37.7% to 54.5%–63.1% while lowering held-out FPR below 1D SPOT (9.77% vs. 10.27%) and cutting delay four- to five-fold (402.7 → 71.8–97.3 steps). Adding a soft temporal and cross-perspective attention pooling gate over the 224 dials yields no further gain over discrete Boolean switches (0.137 vs. 0.138 causal F1), because continuous soft weights re-introduce cross-coordinate dilution that independent per-dial circuit breakers prevent by construction.

D. Horizon Length, Score-Emission Latency, and False-Alarm Control

Target horizon $S = 64$ imposes a 63-step emission lag before an alarm can trigger (Section IV-C). To evaluate detection quality versus latency against baselines at matched emission lag $S-1 \in \{0, 3, 15, 31, 63\}$, Fig. 4(a) retrains models across $S \in \{1, 4, 16, 32, 64\}$ on five streams (three seeds). At $S = 1$

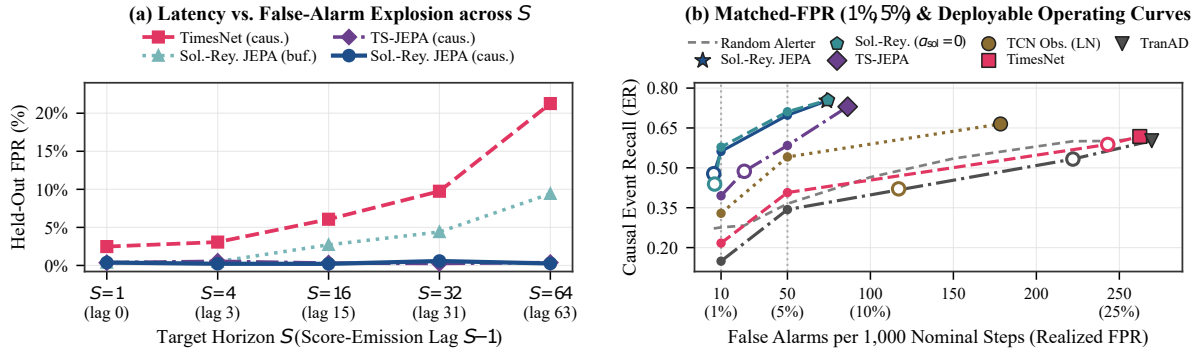


Fig. 4. Horizon length and operating trade-offs. **(a)** Held-out FPR across target horizons $S \in \{1, 4, 16, 32, 64\}$ at matched score-emission latency $S-1 \in \{0, 3, 15, 31, 63\}$ steps (five streams, three seeds). **(b)** Causal event recall versus false alarms across 19 non-GHL streams at equalized test-FPR budgets $f^* \in \{1\%, 5\%\}$ (vertical dotted guides) and deployable calibration quantiles from SPOT $q^*{=}10^{-3}$ (open circles) to $q^*{=}0.05$ (filled markers), plotted against a rate-matched Bernoulli random alerter.

TABLE IV
CONTEXTUAL VS. POINT ANOMALY DETECTION (5 SEEDS; EVENT RECALL (ER)).

	NASA Contextual (n=5 seqs)					NASA Point (n=8)	
(a) Model	**Buf. PR**	**SPOT ER**	**FPR** $f^*{=}.01$	$f^*{=}.05$	$f^*{=}.01$	$f^*{=}.05$	
Soliton-Rey. JEPA	**0.652**	**0.750**	**0.006**	**0.820**	**0.920**	0.683	0.867
Sol.-Rey. ($\alpha_{\text{sol}}{=}0$)	0.641	0.625	**0.006**	**0.820**	**0.920**	0.683	0.867
TS-JEPA	0.644	0.675	0.045	0.720	0.840	0.833	0.867
TimesNet	0.317	0.625	0.415	0.560	0.560	0.750	0.767
TranAD	0.162	0.400	0.392	0.400	0.400	0.767	0.800
TCN Obs. Recon.	0.161	0.375	0.387	0.400	0.400	0.800	0.800
ZScoreMax	0.198	0.375	0.558	0.300	0.300	**1.000**	**1.000**

	19-Stream Contextual ($\lvert z\rvert{\le}5$)			19-Stream Point ($\lvert z\rvert{>}5$)		
(b) Model	$q^*{=}0.05$	$f^*{=}0.01$	$f^*{=}0.05$	$q^*{=}0.05$	$f^*{=}0.01$	$f^*{=}0.05$
Soliton-Rey. JEPA	0.628	0.498	0.590	0.782	0.683	0.781
Sol.-Rey. ($\alpha_{\text{sol}}{=}0$)	**0.640**	**0.540**	**0.603**	**0.785**	0.671	**0.785**
TS-JEPA	0.614	0.520	0.594	0.776	**0.694**	0.765
TimesNet	0.382	0.290	0.301	0.668	0.549	0.567
TranAD	0.402	0.235	0.300	0.640	0.564	0.611
TCN Obs. Recon.	0.472	0.273	0.297	0.639	0.553	0.563
ZScoreMax	0.376	0.288	0.289	0.673	0.594	0.605

Panel (a) stratifies the 10 primary SMAP/MSL streams by NASA's ground-truth taxonomy [4] (contextual: SMAP E-1, A-2, A-7, MSL M-1, and C-1 [2100, 2210]; point: SMAP A-1, P-3, D-3, MSL D-14, D-15, and C-1 [550, 750]). Panel (b) stratifies all $n{=}19$ non-GHL streams into in-range contextual ($\max\lvert z\rvert \le 5\sigma$) vs. out-of-range point ($\max\lvert z\rvert > 5\sigma$) causal events.

(zero lag) and $S = 4$ (3-step lag), TS-JEPA keeps causal FPR at 0.4%–0.6% (vs. 2.5%–3.1% for TimesNet) while maintaining higher causal PR-AUC (0.178–0.189 vs. 0.150–0.162) and short detection delays ($\Delta t = 57.8$–85.2 steps). As S increases from 16 to 64, TimesNet causal FPR climbs from 6.1% to 21.3%, whereas latent predictors remain bounded ($\le 2.0\%$). Consequently, $S = 64$ suits high-rate sensors (≥ 1 Hz, such as 64 Hz Daphnet where 63 steps $= 0.98$ s) or slow subsystem trend monitoring (0.1 Hz SMAP, MSL, and GHL, 10.5 min). Low-rate infrastructure streams ($\le 1\,\text{min}^{-1}$, such as hourly Metro where 63 h lag is prohibitive) and rapid tripwires instead require $S \in \{1, 4\}$.

E. Operating Budgets and Contextual vs. Point Anomalies

Because reconstruction baselines calibrate to 22%–31% realized FPR versus 0.3%–11% for latent predictors, and long segments (402 steps on Daphnet, 1,102 on SMAP) allow a 30% random-alarm detector to touch $> 95\%$ of segments via $1 - (1 - \text{FPR})^L$, we evaluate three comparisons:

1) *Deployable nominal calibration quantiles (q^*, Fig. 4(b), Table II(c)):* At causal calibration quantile $q^* = 0.050$ (set without test labels), TS-JEPA and Soliton-Reynolds JEPA achieve higher event recall (0.730 and 0.752) at lower false-alarm rates than TimesNet (0.618 recall) and TranAD (0.602 recall).
2) *Post-hoc matched realized test-FPR budgets (f^*, Fig. 4(b), Table II(c)):* Equalizing realized test nominal FPR across models (vertical dotted guides in Fig. 4(b)) proves that the JEPA detectors' low false-alarm rate is not an artifact of low alert frequency. At a 1% test FPR budget ($f^* = 0.01$), Soliton-Reynolds JEPA detects 56.2% of anomaly events (57.8% for $\alpha_{\text{sol}}{=}0$) and TS-JEPA detects 39.5%, more than doubling TimesNet (21.7%) and TranAD (14.9%). A similar gap holds at $f^* = 0.05$ (69.8% vs. 34.3%–40.7%), whereas at relaxed budgets ($f^* \ge 0.10$) recall curves converge ($p = 0.625$).
3) *Contextual vs. point anomaly stratification (Table IV):* Out-of-range point anomalies (abrupt amplitude spikes exceeding $\pm 5\sigma$) can be flagged even by a memoryless marginal z-score (`ZScoreMax`), whereas contextual anomalies stay within normal sensor bounds ($\max|z| \le 5\sigma$) and only violate multi-step trajectory structure. In Table IV(a), stratifying the 10 primary SMAP/MSL streams by NASA's official ground-truth taxonomy [4] shows that observation-space detectors collapse on contextual anomalies (0.161–0.317 buffered PR-AUC and 30%–56% recall at $f^* \in \{0.01, 0.05\}$). Soliton-Reynolds JEPA leads every contextual metric, achieving 0.652 buffered PR-AUC, 75.0% causal SPOT recall at 0.6% FPR, and 82.0%–92.0% recall at $f^* \in \{0.01, 0.05\}$ (surpassing TS-JEPA by 8–10 percentage points). Across all 19 non-GHL streams (Table IV(b)), multi-horizon Soliton-Reynolds JEPA and TS-JEPA nearly double the recall of reconstructors on in-range contextual faults ($\approx 60\%$ vs. 30% at $f^* = 0.05$) while also leading on point faults (78.1%–78.5% vs. 56.3%–61.1%).

F. Automated Threshold Calibration Realities on Non-Stationary Telemetry

Static extreme-value thresholding does not achieve its nominal design risk $q = 10^{-3}$ (0.1%) out of sample across all streams and attribution modes. Across all 33 streams (Table V(a)), standard SPOT achieves stream-macro causal FPR of 0.38% for Soliton-Reynolds JEPA and 1.51% for TS-JEPA, compared to 14.23%–16.11% for TranAD and Times-

TABLE V
THRESHOLD CALIBRATION AND DRIFT DECOMPOSITION (q=10^{-3}).

(a) Model	Rule	Causal FPR	Error (Over.)	Buf. FPR	Buf. F1
TS-JEPA	Standard SPOT	0.017 (0.015)	.014 (15×)	0.064 (0.055)	0.139
	Empirical p99.5	0.025 (0.023)	.022 (23×)	0.076 (0.066)	0.140
Sol.-Rey.	Standard SPOT	**0.004 (0.004)**	**.003 (4×)**	**0.055 (0.046)**	0.102
	Empirical p99.5	0.010 (0.009)	.008 (9×)	0.064 (0.054)	0.101
TimesNet	Standard SPOT	0.180 (0.161)	.160 (161×)	0.222 (0.199)	0.140
	Empirical p99.5	0.191 (0.175)	.174 (175×)	0.234 (0.213)	0.141
TranAD	Standard SPOT	0.160 (0.142)	.141 (142×)	0.222 (0.196)	0.094
	Empirical p99.5	0.171 (0.154)	.153 (154×)	0.234 (0.209)	0.096

	Refitted Causal FPR (n=19)			Excess Share (%)		
(b) Model	FPR_{chr}	FPR_{blk}	FPR_{pt}	Inter-Blk	Serial	i.i.d.
TS-JEPA	0.0229	0.0073	0.0033	71.3	18.1	10.5
Soliton-Rey.	**0.0056**	**0.0047**	**0.0020**	**18.7**	59.5	21.8
TimesNet	0.2302	0.0427	0.0035	81.8	17.1	1.1
TranAD	0.2103	0.0464	0.0034	78.3	20.5	1.1

In (a), cell-weighted means (n=33 streams; stream-macro in parentheses); Error is $|FPR_{macro} - q|$ and Over. is FPR_{macro}/q. In (b), 95% CIs for inter-block drift $FPR_{chr} - FPR_{blk}$: TS-JEPA $+.0156$ [$+.001$, $+.034$], Sol.-Rey. $+.0009$ [$-.002$, $+.004$], TimesNet $+.1874$ [$+.078$, $+.305$], TranAD $+.1639$ [$+.072$, $+.267$].

Net. Replacing parametric SPOT with a non-parametric empirical 99.5th percentile ($p_{99.5}$) yields nearly identical results (0.9%–2.3% causal FPR for JEPAs vs. 15.4%–17.5% for reconstructors), demonstrating that reconstructor false-alarm flooding is not an artifact of GPD tail fitting. Under $S = 64$ buffered attribution, however, smearing each window score across 64 consecutive steps dilates every isolated exceedance into a 64-step cluster, inflating buffered FPR to 4.6%–5.5% for JEPAs and nearly 20% for reconstructors under both rules. Static SPOT is therefore near-calibrated for Soliton-Reynolds JEPA only in strictly causal mode. Two diagnostics explain why causal thresholds hold on latent residuals:

1) *Three-stage refitted permutation decomposition (Table V(b)):* To separate inter-block regime drift from serial correlation and GPD tail-fit error, we pool nominal calibration and evaluation scores $\mathbf{v}_{\text{pool}} = [v_{\text{cal}}, v_{\text{hold}}]$ across all 19 non-GHL streams and *refit* SPOT ($u = p_{98}, q = 10^{-3}$) under three splits: chronological (FPR_{chr}), $L = 64$ block permutation (FPR_{blk}, removing macro-block drift while preserving intra-block serial correlation), and pointwise i.i.d. permutation (FPR_{pt}). Under i.i.d. shuffling, every model achieves low false-alarm rates near the 0.1% target ($FPR_{pt} \leq 0.35\%$). In chronological deployment, inter-block drift ($FPR_{chr} - FPR_{blk}$) accounts for 71.3% of excess causal FPR in single-horizon TS-JEPA and 78.3%–81.8% in reconstructors, whereas in barycenter-centered Soliton-Reynolds JEPA the inter-block drift share drops four-fold to 18.7% ($FPR_{chr} = 0.56\%$ vs. $FPR_{blk} = 0.47\%$, Table V(b)).
2) *GPD tail stability, declustering, and rolling adaptation:* On the 19 non-GHL streams at $u = p_{98}$, GPD fits converge with 0% solver fallback and low threshold uncertainty ($\text{SE}(\tau)/\tau \leq 3.5\%$). Grimshaw GPD fits hit the lower shape boundary ($\xi = -0.90$) in only 4.4% of JEPA cells versus 34.4%–54.4% for reconstructors. Applying Ferro–Segers runs declustering ($r = 5$) [39] prior to GPD fitting trims stationary causal FPR to 0.20% for Soliton-Reynolds JEPA and 0.44% for TS-JEPA. Under discrete step, variance, and regime shifts across 16 streams, adaptive rolling quantiles hold causal TS-JEPA FPR below 0.40% (vs. 0.8%–3.0% for TimesNet), whereas continuous monotonic mean ramps defeat trailing rolling windows ($> 32\%$ FPR across all models).

VIII. CONCLUSION AND FUTURE WORK

We evaluated how Joint-Embedding Predictive Architectures condition anomaly residuals under automated calibration across 33 primary and 141 archive-wide continuous telemetry streams:

1) *Factorial Mechanism Isolation and Multi-Horizon Whitening:* Matched-encoder ablations separate score bounding from representation learning: LayerNorm bottlenecks compress residual kurtosis (γ_2 from 6.10 to 0.91), learned latent projections cut calibration drift in half (D_{KS} from 0.376 to 0.214 in TS-JEPA), and multi-horizon barycenter whitening in Soliton-Reynolds JEPA lowers drift further to 0.179 at 0.6% causal FPR. On the fixed backbone, post-hoc Lieb–Thirring soliton modulation is statistically equivalent to the unmodulated multi-horizon envelope under TOST ($p_{\text{TOST}} < 10^{-4}$), and short horizons ($S \in \{1, 4\}$) eliminate the 63-step emission lag.
2) *224-Bit Multi-Perspective Circuit Breakers and Archive-Wide Generalization:* Latent scoring lowers held-out false alarms three- to four-fold under buffered attribution and ten- to forty-fold causally while doubling reconstructor event recall at a 1% test FPR budget (56.2% vs. 21.7% for TimesNet). Replacing 1D scalar score collapse and SPOT extrapolation with a 224-bit fluid-dynamic circuit-breaker panel and 32 fast right-edge onset relays eliminates hijacker-dial threshold inflation, more than doubling causal Point-F1 ($0.052 \rightarrow 0.138$ on n=19 streams, $p = 1.5{\times}10^{-5}$; 0.067–0.077 $\rightarrow 0.139$ across all 141 streams, $p = 1.2{\times}10^{-16}$), lifting causal event recall to 72.7%, and cutting detection delay in half.
3) *Calibration Drift and Adaptive Thresholding:* Static SPOT overshoots $q = 10^{-3}$ under buffered window smearing and by $15\times$ causally for single-horizon TS-JEPA, whereas barycenter-whitened Soliton-Reynolds JEPA reduces causal overshoot to 0.38% stream-macro FPR across 33 streams and cuts the inter-block drift share four-fold (71.3% to 18.7%). Runs-declustered SPOT, per-dial hijack guards, and rolling quantiles narrow the remaining gap across step and variance shifts, motivating regime-conditioned conformal calibration.

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